Design Plan for Package R: 'BFRdist'

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1. Description and Data Requirements

The `BFRdist` package provides computational and visual tools for the novel BFR distribution, designed specifically for Generalized Radar Vegetation Index (GRVI) analysis. The package translates this theoretical model into a practical tool by implementing the standard `d/p/q/r` family of functions, parameter estimation, and visualization utilities.

Data Requirements: Users will primarily input empirical GRVI datasets or generate synthetic data to fit the BFR distribution. The package requires numeric vectors of observations. Because the BFR distribution is defined strictly on the unit interval, all input data (such as empirical GRVI datasets) must be strictly greater than 0 and less than 1. Missing values (`NA`) and infinite values are not permitted.

GitHub Repo:

https://github.com/luodani272/DATA501

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2. Mathematical Formulation

The core functions of the BFR distribution, as derived by Bhattacharya, Frery, and Rey (2026), are mathematically defined as follows:

Probability Density Function (PDF)

$$f_{BFR}(x;\alpha,\beta) = \frac{2\beta}{\pi} \cdot \frac{\alpha^{\beta} x^{\beta-1}(1-x)^{\beta-1}}{\alpha^{2\beta}x^{2\beta} + (1-x)^{2\beta}} 1_{(0,1)}(x)$$

where $\alpha > 0$ and $\beta > 0$.

Cumulative Distribution Function (CDF)

$$F_{BFR}(q;\alpha,\beta) = \begin{cases} 0 & \text{if } q \leq 0, \\ \frac{2}{\pi}\arctan\left(\frac{\alpha q}{1-q}\right)^{\beta} & \text{if } 0 < q < 1, \text{ and} \\ 1 & \text{if } q \geq 1, \end{cases}$$

where $\alpha > 0$ and $\beta > 0$.

Quantile Function

$$Q_{BFR}(p;\alpha,\beta) = \left[1 + \alpha\left(\tan\left(\frac{\pi p}{2}\right)\right)^{\frac{-1}{\beta}}\right]^{-1} 1_{(0,1)}(p)$$

where $\alpha > 0$ and $\beta > 0$.

Random Generation Function First generate $u \sim \text{Uniform}(0, 1)$. Then compute:

$$X = Q_{BFR}(u; \alpha, \beta) = \left[1 + \alpha \left(\tan\left(\frac{\pi u}{2}\right)\right)^{\frac{-1}{\beta}}\right]^{-1},$$

and therefore, $X \sim BFR(\alpha, \beta)$.

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3. Object-Oriented (S3) Implementation

To leverage the object-oriented features of R, the primary estimation function `fit_bfr()` will return a custom S3 object of class `bfr`. We will implement the following S3 generic methods to interact with this object:

- **`print.bfr`**: To display basic fitted parameter estimates and the estimation method used.
- **`summary.bfr`**: To output standard errors and goodness-of-fit metrics (Log-Likelihood, AIC, and BIC).
- **`plot.bfr`**: To generate Pearson-like SKG/KUM diagrams automatically using `ggplot2`.
- **`predict.bfr`**: To simulate new predictions based on the fitted model.

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4. Ethical and Legal Dimensions

The `BFRdist` package is a statistical and computational tool designed primarily to analyze environmental and satellite data (GRVI datasets). Because it deals with remote sensing data regarding vegetation, there are no direct ethical concerns related to human subjects, privacy, or personally identifiable information. Users of the package must ensure that any proprietary GRVI data fed into the package is obtained legally and cited properly.

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5. Unit Testing Plan

To verify the package's validity, a comprehensive testing suite will be implemented for the main mathematical functions. Test frameworks that will be used include `testthat` as well as `Rcpp`. These unit tests will be stored in a folder on the same folder level as /src, named /tests.

- **Core Function Integrity:** Unit tests will verify that `dbfr`, `pbfr`, `qbfr`, and `rbfr` return mathematically valid outputs (e.g., CDF bounds are exactly 0 to 1, and densities are non-negative). These tests will leverage `testthat`, and due to the simplicity of said functions can be implemented all as one method.
- **Estimator Accuracy:** C++ quantile-based estimators (`estimateBFR_cpp`) will be tested against known parameters from synthetic data to ensure accurate recovery of α and β. Unit testing for the C++ portion of the quantile estimator will leverage `Rcpp` while the R portion will also use `testthat`.

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6. Deliverables and Timeline

- **Week 7:** Design Plan submission.
- **Week 8:** Base Estimators (ML & Moments) and plot function completed.
- **Week 10:** Draft Package and Testing Plan. This will be a working, frozen package version submitted for peer review, including core functions, a readable vignette, and a reproducible test script.
- **Week 12:** Peer Review Report execution.
- **Week 13/14:** Finalized package delivery and presentation.

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7. Software Architecture Diagrams

The software architecture is visualized in Figure 1 and Figure 2.

- **Figure 1 (Dataflow Diagram):** Illustrates how raw GRVI data enters the `fit_bfr()` process, cycles through the C++ back-end and core functions for estimation, and outputs an S3 `bfr` object. This object subsequently feeds into the summary statistics and S3 methods.

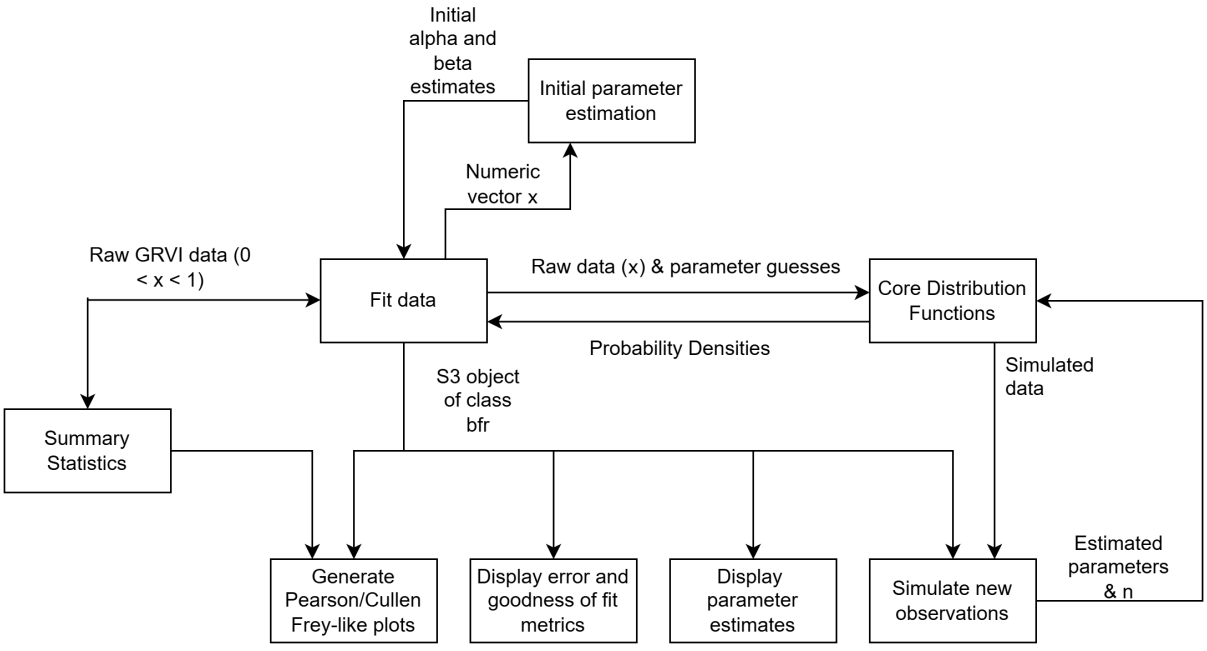


Figure 1: Dataflow Diagram

- **Figure 2 (Sequence Diagram):** Details the chronological execution of functions, mapping the interactions between the R Environment, the C++ back-end (`estimateBFR_cpp`), the core distribution functions, and the S3 methods.

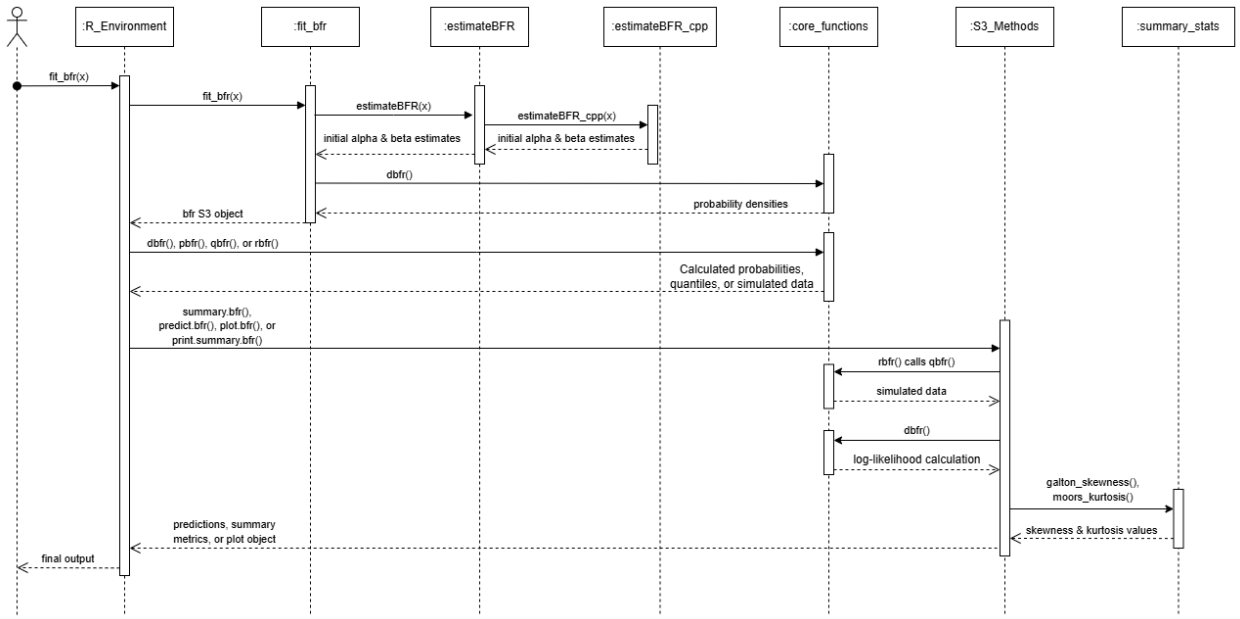


Figure 2: Sequence Diagram

References

- Bhattacharya, A., Frery, A. C., & Rey, A. A. (2026). A Statistical Framework for Bounded Measurements with Applications to SAR Data. *IEEE Transactions on Geoscience and Remote Sensing*.